

Variance

Math 252

Warm-up: Expected Value

1. Consider the random variable described below.

value	0	1	5	50
probability	???	10/100	5/100	1/100

- a. Fill in the missing probability.
 - b. Compute the expected value.
 - c. Suppose these are the values and probabilities of prizes given out in a raffle. If the people running the raffle charge \$1 per ticket, how much do they expect to make if they sell 1000 tickets?
2. What does $E(X)$ tell us about X ?

Variance

Variance is a measure of how variable or how “spread out” a random variable is. We will determine the variance by looking at the difference between X and $E(X)$. If this difference tends to be large, then there is more variability. If it tends to be small, there is less.

Here’s how not to do this:

3. Let X be a any random variable. What can you say about $E(X - E(X))$? [Hint: use facts about expected value to keep this simple.]

Definition

To fix this problem, we need to do something the differences $X - E(X)$. We could take absolute values, but it works better to square the differences. So here is our definition of variance.

$$\text{Var}(X) = E((X - \mu_X)^2) = E((X - E(X))^2)$$

4. Use this definition to compute the variance of the random variable in problem 1.
5. Compute the variance of a $\text{Binom}(1, p)$ random variable.
6. Compute the variance of a $\text{Binom}(2, 1/2)$ random variable.
7. Compute the variance of a $\text{Binom}(2, 1/4)$ random variable.

Alternate notation

Just like we can denote $E(X)$ with μ_X , we have another notation for variance as well: σ_X^2 . The square root of the variance, denoted $\text{SD}(X)$ or σ_X , is called the **standard deviation**.

Handy short cut

$$\text{Var}(X) = E(X^2) - E(X)^2$$

8. Use the short-cut to compute the variance of the random variable in problem 1.
9. Use the short-cut to compute the variance of the random variable in problem 5.
10. Use the short-cut to compute the variance of the random variable in problem 6.

Linear Transformations

9. Prove the following handy rules for variance.
 - a. $\text{Var}(aX) = a^2 \text{Var}(X)$
 - b. $\text{Var}(X + b) = \text{Var}(X)$
10. What are the equivalent rules for standard deviation?
11. If $E(X) = 10$ and $\text{Var}(X) = 2$, what are the mean and variance of $2X - 3$?

Sums and Variance

There is a sum rule for variance, but it only applies if the random variables are **independent**.

Two random variables are **independent** if and only if

$$P(X = a \wedge Y = b) = P(X = a) \cdot P(Y = b) \text{ for every } a \text{ and } b.$$

If X and Y are independent, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

12. If X and Y are independent, $E(X) = 10$, $\text{Var}(X) = 4$, $E(Y) = 20$, and $\text{Var}(Y) = 9$, what are the mean and variance of $X + Y$ and $X - Y$?
13. Use the sum rule for variances to compute the variance of a $\text{Binom}(2, p)$ random variable.
14. Use the sum rule for variances to compute the variance of a $\text{Binom}(n, p)$ random variable.
15. What is the sum rule for standard deviation?